AICTE VERIFIED STUDENT REGISTER

Synthetic Demo Data and Test Scenarios

Safe fake data conventions, CSV shape, role seed accounts and repeatable end-to-end demonstrations.

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Audience: AI coding agent + project owner

Document-control note
This is an implementation specification for a synthetic-data prototype, not an official AICTE standard, production approval, legal opinion, or proof of live government integration.

Source discipline: requirements from the PS 26263 implementation report and Sigma Batch curriculum are distinguished from proposed engineering choices. Any item that depends on a real policy, approval, API, credential trust chain, or deployment environment must be confirmed rather than guessed.

1. Why this document exists

The coding agent needs repeatable data to build and test each screen. This specification defines the shape and meaning of fake demo records. It does not contain real students, real government IDs, real accounts or official quota limits.

No real personal data
All names, identifiers, phone tokens, colleges and records in this specification are fabricated for software testing. Never replace them with real APAAR/Aadhaar values or student records.

2. CSV upload contract

Column	Required?	Meaning / validation
studentCode	Yes	Synthetic local code such as STU-001. It is a demo reference, not an official AICTE identifier.
studentName	Yes	Synthetic display name; trim whitespace; empty name is L1 ERROR.
courseCode	Yes	Must match a course configured for the authenticated institution and academic year.
academicYear	Yes	Must match the batch academic year and approved demo configuration.
bucketCode	Yes	Must match a configured bucket code. The template must derive permitted values from active demo config.
externalIdDemo	Required only when the mock identity check is configured for this test	Fake external identifier for mock-only format checking. A 12-digit format fixture mirrors the report’s example but does not validate a real identity.
contactToken	No, used in selected risk fixtures	Synthetic token such as PHONE-DEMO-A; never a real phone number.
yearOfStudy	Yes if the configured rule set requires it	Integer or permitted enum; exact constraints must come from the selected demo rule set.
admittedOn	Yes if the configured rule set requires it	ISO date YYYY-MM-DD; must be consistent with demo cutoff configuration.

The institutionId and submitting user must come from the authenticated session and stored user record. Do not accept them from a CSV as trusted authority. The client may perform pre-checks, but server validation is authoritative.

3. Clean sample CSV

Save the following as clean_batch.csv for the first successful workflow. The values are fabricated. Keep the selected seat capacities high enough for this clean fixture.

```
studentCode,studentName,courseCode,academicYear,bucketCode,externalIdDemo,contactToken,yearOfStudy,admittedOn
STU-001,Aarav Sharma,CSE,2026-27,REGULAR,111111111111,PHONE-DEMO-A,1,2026-08-01
STU-002,Neha Verma,CSE,2026-27,REGULAR,222222222222,PHONE-DEMO-B,1,2026-08-02
STU-003,Kabir Singh,CSE,2026-27,REGULAR,444444444444,PHONE-DEMO-C,1,2026-08-03
STU-004,Meera Das,CSE,2026-27,REGULAR,555555555555,PHONE-DEMO-D,1,2026-08-04
```

4. Negative fixtures

Fixture	Constructed condition	Expected system behaviour
missing_required_field.csv	A row has an empty courseCode or studentName.	Create L1 ERROR with row number and field name; do not commit.
bad_mock_external_id.csv	externalIdDemo is 11 digits while the selected demo format rule requires 12.	Return mock-format L2 ERROR. Do not claim a real registry lookup occurred.
duplicate_in_batch.csv	The same studentCode/externalIdDemo and course/year are repeated in a single batch.	L3 ERROR references both row numbers; do not merge or delete silently.
bucket_overflow.csv	Demo rule config capacity for NRI is 3; fixture tries to reserve 4 NRI seats.	L4 ERROR; no partial commit and no seat counter above 3.
registry_pending.csv	Mock adapter is configured to return SERVICE_UNAVAILABLE/PENDING.	Set the identity check to PENDING; never report verified; batch decision follows explicit pending policy.
cross_college_conflict.json	STU-005 has apparent active enrolments at College A and College B for the same academic year.	Create L5 conflict case; leave case open until an authorised state transition/justification resolves it.
shared_contact_risk.json	Twelve synthetic student records share contactToken PHONE-DEMO-SHARED.	Create an L7 risk signal with count/evidence; do not block solely for this signal or label fraud.
stale_policy_batch.json	Batch was validated under DEMO-v1; current active demo rules become DEMO-v2 before certification.	Revalidate or reject according to the documented policy-version rule; never silently use mixed capacities.

5. Seat-ledger demo configuration

Use a separate synthetic configuration file and display its demo label in the admin UI. The following capacities are for a deterministic test only; do not ship them as actual AICTE limits.

```
{
  "academicYear": "2026-27",
  "policyVersion": "DEMO-v1",
  "sourceType": "SYNTHETIC_DEMO_CONFIG",
  "institutionCode": "ABC-DEMO",
  "courseCode": "CSE",
  "buckets": [
    { "code": "REGULAR", "capacity": 57, "countsWithinIntake": true },
    { "code": "NRI", "capacity": 3, "countsWithinIntake": true },
    { "code": "TFW", "capacity": 3, "countsWithinIntake": false },
    { "code": "INTERNATIONAL", "capacity": 3, "countsWithinIntake": false }
  ]
}
```

6. Seeded demo accounts

Create one synthetic account per role. Use reserved .test email addresses and synthetic institution/region references. Do not place fixed working passwords in this PDF, source code, or version control. The seed script should require locally supplied development secrets or generate one-time random passwords and display them only in the developer terminal.

Account alias	Role	Assigned scope in demo
operator.demo@example.test	COLLEGE_OPERATOR	ABC-DEMO institution
head.demo@example.test	COLLEGE_HEAD	ABC-DEMO institution
national.demo@example.test	AICTE_NATIONAL_OFFICER	National aggregate scope
regional.demo@example.test	AICTE_REGIONAL_OFFICER	REGION-DEMO only
admin.demo@example.test	AICTE_SYSTEM_ADMIN	System user/config management
issuer.demo@example.test	DEGREE_AWARDING_BODY	ISSUER-DEMO scope only
verifier.demo@example.test	VERIFIER	VERIFIER-DEMO organisation

Account alias	Role	Assigned scope in demo
student.demo@example.test	STUDENT	STU-001 only

Seed script acceptance: refuses to run in production mode; hashes passwords; does not print hashes; is safe to rerun (upserts predictable demo data without duplicating users); creates no real data; warns that accounts are demo-only.

7. Repeatable seven-minute demo flow

Step	Action	Expected outcome
1. Upload	Login as operator and upload clean/negative fixture.	Batch receives ID/status; progress and row count appear.
2. Fix errors	Open issue grid, correct allowed fields and revalidate.	Blocking issues clear only when inputs actually satisfy rules.
3. Seat ledger	Run capacity fixture using DEMO-v1 config.	Within-capacity categories pass; an NRI capacity of 3 rejects a fourth reservation.
4. Conflict	Open the STU-005 conflict fixture.	Both authorised college sides see appropriate case; transition resolves only after valid action.
5. Risk queue	Open shared-contact risk signal.	Score/reasons/evidence appear; a reviewer can mark reviewed or false positive.
6. Consent	Login as verifier, submit request, then as student approve it.	Verifier sees only approved fields and a recorded access event.
7. QR + audit	Show demo QR status, revoke it, then run audit-chain test.	Status changes to revoked; controlled audit alteration fixture is detected.

8. Test fixture hygiene

- Keep fixtures deterministic: fixed synthetic IDs, dates and rule versions. Do not use random values that make expected test outputs unstable.
- Use separate fixtures for different failure modes so one early blocking error does not mask another test.
- Reset demo data only through a documented development-only reset command; never add a public reset endpoint.
- Mark all UI labels and export filenames as DEMO where confusion with real credentials is possible.
- Capture the exact configuration, command and test output when making any performance statement.

9. Open decisions

- The owner must approve exact column names before the first implementation contract is frozen.
- A real identity registry contract is not included; mock outcomes are selected by deterministic fixtures.
- Official seat rules remain TBD. DEMO-v1 capacities are synthetic tests only.
- Any legal basis for non-consensual verifier access is out of prototype scope; default test flow requires student approval.